Academic Year	Module	Assessment Number	Assessment Type
2026	6CS012 Artificial Intelligence and Machine Learning	Part 1	QnA

Question and Answer

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Long Question

Question:

1. You are a Data Scientist at a leading digital payment platform such as eSewa, working on large-scale user transaction data.

- Identify two high-impact business problems where unsupervised learning can provide value (e.g.,customer segmentation, fraud detection, behavioral profiling).
- For each problem:
  - Clearly define the problem in a real-world context.
  - Propose a suitable unsupervised learning approach (e.g., clustering, anomaly detection, dimensionality reduction) and recommend specific algorithms (e.g., K-Means, DBSCAN, Autoencoders).
  - Discuss how the model outputs can be integrated into real-time or batch systems to support business decisions.
  - Briefly mention one limitation or challenge of your proposed approach.
- Assume limited labeled data and large-scale transactions.

Answer:

The digital payment system eSewa uses unsupervised learning methods to solve its first problem which involves detecting fraudulent transactions.

The world produces millions of transactions each day but some of these transactions involve fraudulent activities which include account takeover and card testing. The existing manual systems which operate based on fixed rules fail to detect new fraudulent activities which were not present in their existing detection patterns.

The system uses Isolation Forest and Autoencoders for its anomaly detection method. The models develop understanding of typical transaction behavior which includes four factors: amount and location and device and time.

The system operates through two ways: it processes historical transactions in batch mode during nighttime and it uses an automatic scoring system through an API which operates on edge servers with a small autoencoder model. The system uses two-factor authentication and temporary account suspension as security measures which activate when alerts occur.

The system generates excessive false positive results which create frustration for real users while it needs ongoing model updates to keep up with changing patterns of fraud.

Problem 2: Customer Segmentation for Targeted Offers

eSewa plans to create user groups based on their spending patterns which include users who pay bills and users who shop with small amounts.

Our method uses K-Means and GMM clustering to analyze transaction data through its frequency and average spending and category distribution. The data becomes more manageable through PCA-based dimensionality reduction.

The system updates user segments through weekly batch processing which sends updates to a recommendation system that activates push notifications and in-app promotions.

The system requires ongoing re-clustering and business validation because its clusters become unstable after some time.

Short Questions

1.Overfitting

Question:

Overfitting is a common challenge in deep learning, especially when models have high capacity.

- Describe at least two techniques used to reduce overfitting (e.g., dropout, early stopping, data augmentation, L2 regularization).
- For each technique:
    - Explain the intuition behind how it works.
    - Describe how it affects the training process or model behavior.
    - Provide a real-world application example (e.g., image classification, text classification, fraud detection).

Answer:

Two techniques to reduce overfitting:

- Dropout
  The technique functions by applying random neuron zeroing to the training process which stops features from building co-adaptive relationships.

  The network requires learning multiple ways to represent information because the system operates like an ensemble that consists of many smaller networks.

  The implementation of dropout after fully connected layers with a rate of 0.5 in image classification using CIFAR-10 leads to a test error reduction of approximately 5 percent.

- Early Stopping
  The system requires validation loss monitoring throughout training until it shows no improvements for a specific number of training periods at which point training terminates.

  The training process stops the model from learning the unique noise patterns which exist only in the training dataset.

  The implementation of early stopping with a patience setting of three in text sentiment analysis prevents the model from overfitting to outlier samples in the training corpus.

2. Neural Network Architecture

Question:

1. Explain the fundamental differences between a Convolutional Neural Network (CNN) and Recurrent Neural Network (RNN):

- In terms of:
  - Input structure and data type
  - Information flow and architecture design
  - Parameter sharing mechanism
- Provide suitable real-world examples where each architecture is preferred (e.g., image classification for CNNs, sequential/time-series modeling for RNNs).
- Briefly discuss common challenges encountered during training deep learning models, such as:
  - Vanishing or exploding gradients
  - Overfitting
- For each challenge, mention at least one mitigation technique (e.g., batch normalization, gradient clipping, early stopping).

Answer:

An RNN is a class of neural networks which processes sequential data because it requires knowledge of input sequence order. RNNs differ from feedforward networks because they use recurrent connections to maintain hidden states which transmit information across different time steps. The CNN architecture functions as a specialized neural network framework which employs convolutional operations to learn spatial feature hierarchies from grid-based data.

Aspect	**CNN**	**RNN**
Input structure	Grid-like data (images, 2D matrices)	Sequential data (time series, text)

Architecture	Convolution + pooling layers; local connectivity	Recurrent cells with hidden state; temporal loops
Parameter sharing	Kernels shared across spatial locations	Weights shared across time steps

The first example demonstrates CNN through image classification which uses ResNet to classify images from the ImageNet dataset. The second example demonstrates RNN through language modelling which uses LSTM to predict the next word in a sequence.

The training challenges for this project include vanishing and exploding gradients which require the implementation of gradient clipping and batch normalization to solve the problem. The research project faces two main challenges which include overfitting and multiple solutions which researchers can use to solve this problem.